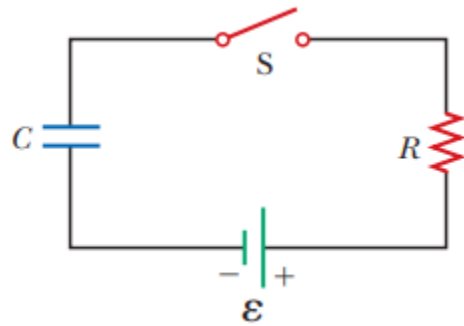


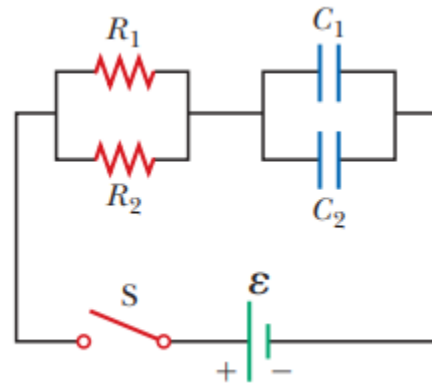
Sheet 6

Transient Circuit Analysis

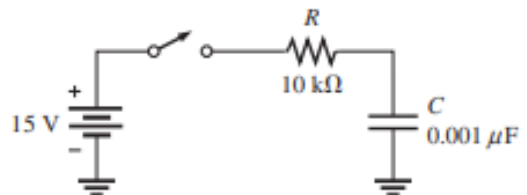
1. Consider a series RC circuit as in the Figure for which $R = 1\text{ MV}$, $C = 5\text{ }\mu\text{F}$, and $\varepsilon = 30\text{ V}$. **Find** (a) the time constant of the circuit and (b) the maximum charge on the capacitor after the switch is thrown closed. (c) **Find** the current in the resistor 10 s after the switch is closed.



2. The circuit in the Figure contains two resistors, $R_1 = 2\text{ k}\Omega$ and $R_2 = 3\text{ k}\Omega$, and two capacitors, $C_1 = 2\text{ }\mu\text{F}$ and $C_2 = 3\text{ }\mu\text{F}$, connected to a battery with $\varepsilon = 120\text{ V}$. If there are no charges on the capacitors before switch S is closed, **determine** the charges on capacitors (a) C_1 and (b) C_2 as functions of time, after the switch is closed.

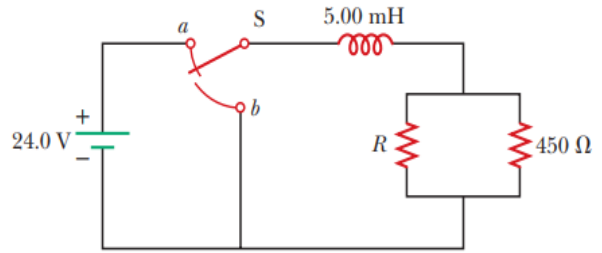


3. In the circuit of the Figure, the capacitor initially is uncharged. **Determine** the capacitor voltage at the following times after the switch is closed: (a) $10\text{ }\mu\text{s}$ (b) $20\text{ }\mu\text{s}$ (c) $30\text{ }\mu\text{s}$ (d) $40\text{ }\mu\text{s}$ (e) $50\text{ }\mu\text{s}$

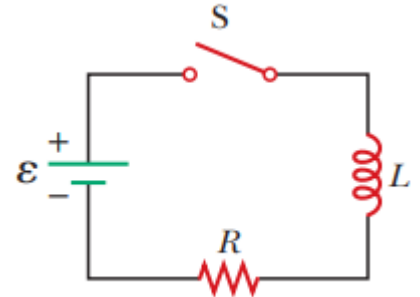


4. Consider the circuit shown in the Figure.

(a) When the switch is in position a, for what value of R will the circuit have a time constant of 15 ms ? (b) What is the current in the inductor at the instant the switch is thrown to position b?



5. In the circuit shown in the Figure, let $L = 7\text{ H}$, $R = 9\text{ V}$, and $\varepsilon = 120\text{ V}$. What is the self-induced emf 0.2 s after the switch is closed?



6. Consider the circuit in the Figure of the previous problems, taking $\varepsilon = 6\text{ V}$, $L = 8\text{ mH}$, and $R = 4\text{ V}$. (a) what is the inductive time constant of the circuit? (b) Calculate the current in the circuit 250 ms after the switch is closed. (c) What is the value of the final steady-state current? (d) After what time interval does the current reach 80% of its maximum value?
7. A 140 mH inductor and a 4.9 V resistor are connected with a switch to a 6 V battery as shown in the Figure. (a) After the switch is first thrown to a (connecting the battery), what time interval elapses before the current reaches 220 mA ? (b) What is the current in the inductor 10 s after the switch is closed? (c) Now the switch is quickly thrown from a to b. What time interval elapses before the current in the inductor falls to 160 mA ?

